ANLP Project Mid Submission: Generating and Evaluating 3D Relevancy Maps for Text Queries

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Abstract

This project explores the task of grounding natural language queries in 3D scenes through the generation of 3D relevancy maps. Initially, we aimed to develop a new Vision-Language Model (VLM) based approach for 3D grounding, but the final implementation focuses on a simplified and practical pipeline. The approach uses **LLaVA** to extract key descriptive keywords from free-form natural language queries, which are then used as input to **LERF** (Language Embedded Radiance Fields) for producing 3D relevancy maps.

Beyond implementation, the project conducts an extensive comparative **probing-based analysis** of **LERF** and **ConceptFusion**, evaluating their strengths and limitations across different categories of linguistic complexity — from simple object queries to spatial and abstract reasoning. The results highlight the contrasting behaviors of these architectures: while both models perform well on concrete object localization, they diverge in their handling of functional affordances and abstract concepts. This analysis provides valuable insights into the representational capabilities and linguistic grounding limits of current open-vocabulary 3D understanding systems.

1 Introduction

1.1 Project Overview

Grounding natural language in 3D environments is a key challenge for advancing embodied AI and immersive computing systems. A robust 3D grounding system should be able to interpret open-ended text queries and generate corresponding *3D relevancy maps*—continuous fields that indicate how strongly each region in a 3D scene corresponds to the meaning of the given query.

In this project, we explore a simplified and interpretable approach to this problem. Our system first employs the **LLaVA** (Large Language and Vision Assistant) model to extract concise and meaningful *keywords* from natural language queries. These extracted terms are then used with the **LERF** framework to generate 3D relevancy maps. This modular pipeline allows us to extend LERF's original capability—limited to object-level queries—to handle more general or instruction-like queries without modifying the model's architecture.

However, the major focus of the project after the mid-submission phase has been on a comprehensive **comparative analysis** of existing 3D grounding methods—particularly **LERF** and **ConceptFusion**. Using a probing-based evaluation framework, both models were tested with structured query sets spanning varying linguistic complexities. The analysis reveals their different strengths: ConceptFusion tends to generalize better for affordance-based reasoning, while LERF exhibits brittleness to phrasing and poor handling of spatial and abstract queries. These findings offer deeper insight into the semantic grounding behavior of each architecture.

1.2 Objectives

The objectives of the final phase of this project are as follows:

- To design and implement a simple and modular LLaVA–LERF pipeline that maps general natural language queries into relevant object-based inputs for LERF, enabling relevancy map generation without retraining.

- To perform a systematic, probing-based comparative study of **LERF** and **ConceptFusion** models, analyzing their performance across categories of increasing linguistic and conceptual complexity.

- To identify the strengths, weaknesses, and representational boundaries of these models, particularly regarding their handling of fine-grained attributes, spatial relations, and abstract reasoning.

The outcomes of this analysis provide a clearer understanding of the current landscape of open-vocabulary 3D scene understanding and set the stage for future research that integrates semantic reasoning with geometric grounding.

2 Related Work

The task of grounding language in 3D scenes has seen the emergence of several distinct architectural paradigms, broadly categorized into implicit field approaches, explicit mapping approaches, and high-level reasoning frameworks.

Language Embedded Radiance Fields (LERF)(Kerr et al., 2023) pioneer an approach that directly embeds language into an implicit neural representation of a scene. LERF augments a standard NeRF(Mildenhall et al., 2020) by adding a second MLP head that predicts a CLIP feature vector for any 3D point

$\mathbf{x}$ at a given scale s. The model is trained by jointly optimizing the standard NeRF color reconstruction loss and a language loss, which is supervised by volume-rendered CLIP(Radford et al., 2021) embeddings against 2D CLIP features from training views. This effectively "distills" multi-view 2D semantic information into a dense, 3D-consistent language field. A key innovation is the use of a multi-scale feature pyramid, allowing a single 3D point to be associated with concepts at different scales (e.g., "wooden spoon" vs. "cutlery"). To enforce geometric consistency and produce crisper maps, LERF also incorporates a regularization field supervised by features from the DINO model(Caron et al., 2021), which is known for its excellent emergent object segmentation properties. After training, a LERF can be queried efficiently by computing the cosine similarity between a text prompt's embedding and the language field's output across the entire 3D volume.

ConceptFusion(Jatavallabhula et al., 2023) takes a fundamentally different, decoupled approach. It leverages existing technologies, first using a standard SLAM system to build an explicit 3D point cloud map from an RGB-D video stream. In parallel, it extracts dense, pixel-aligned semantic features from each 2D frame using a foundation model like CLIP. The core innovation is its method for

zero-shot pixel alignment, which generates pixel-level features from global-level models like CLIP. It uses a class-agnostic instance segmentation model to identify regions, computes "local" CLIP features for each region and a "global" feature for the whole image, and then intelligently fuses them to produce a final feature vector for each pixel. These dense 2D feature maps are then "lifted" into 3D by projecting them onto the SLAM-generated geometry using known camera poses and depth. Features from multiple views are fused over time to create a robust, unified 3D semantic map where each point is annotated with a rich feature vector.

While LERF and ConceptFusion focus on dense relevancy maps, other models aim for higher-level reasoning. LLaVA (Large Language and Vision Assistant)(Liu et al., 2023) is a pioneering multimodal conversational agent that connects a pre-trained CLIP vision encoder to a Large Language Model (LLM) like Vicuna via a simple projection matrix. It is fine-tuned on a large dataset of instruction-following examples generated by GPT-4 from 2D images, enabling it to hold coherent conversations about visual content but without any innate 3D understanding. SceneGPT(Chandhok, 2024) bridges this gap by introducing an abstraction layer. It first processes a 3D scene into a structured, language-readable

3D Scene Graph (3DSG), typically a JSON object describing objects, their properties, and spatial relationships. This structured text is then passed into an LLM's context window, allowing it to parse the representation and answer complex spatial and functional queries. SceneGPT(Chandhok, 2024) excels at high-level reasoning but does not produce a dense relevancy map.

The distinct philosophies of these models lead to a trade-off between dense, low-level grounding and sparse, high-level reasoning. The progression from LERF/ConceptFusion to LLaVA/SceneGPT illustrates an evolution from localization to understanding. The former systems answer "Where is concept X?", while the latter answer "What about concept X?". This suggests a powerful future architecture would likely integrate both capabilities, using a dense grounding system as a "perceptual front-end" for an LLM-based "reasoning back-end."

3 Dataset Details

We evaluate our models using three sources of 3D scene data: the LERF dataset, the LERF-TOGO dataset, and a custom dataset constructed from CAD models using BlenderNeRF. This combination enables testing open-vocabulary object localization, task-oriented grasping, and controlled evaluation of spatial reasoning.

3.1 LERF Dataset

Figure 1: Sample scene from the LERF (Kerr et al., 2023) dataset showing a variety of household objects.

The LERF dataset contains scenes with different types of objects, supporting open-vocabulary queries and instance-level semantic localization. It allows testing the model's ability to identify and localize multiple objects in complex arrangements.

3.2 Custom Dataset Pipeline

Figure 2: Top-down view of a NeRF-reconstructed point cloud from our custom dataset.

We construct our custom dataset using publicly available CAD models, such as apartment floor plans, rendered with BlenderNeRF(Eraafat, 2023) . Multi-view RGB images are generated from predefined camera poses, and the corresponding extrinsic and intrinsic camera parameters are stored in a format compatible with LERF. This setup allows systematic design of scenes with diverse spatial arrangements, placing objects above, below, or beside each other, and enables targeted queries to evaluate spatial reasoning, relative placement, and compositional understanding beyond simple object retrieval.

4 Proposed Methodology

Earlier, we had proposed an approach for the task of getting relevancy maps for general text queries that uses LERF as the main model. Basically, our contribution is to map general text queries to text keywords, which can be directly used for LERF.(Kerr et al., 2023) We have completed this approach by implementing LLAVA (Liu et al., 2023) model.

4.1 Experimental Protocol for Comparative Probing

To rigorously evaluate the comparative performance of the two state-of-the-art frameworks, **LERF** and **ConceptFusion**, a comprehensive and fair experimental protocol is designed. This protocol systematically probes the models' capabilities across a wide range of linguistic complexities on a consistent set of 3D scenes.

The foundation of a fair comparison lies in the correct preparation and deployment of both baseline models. The official open-source implementations of LERF and ConceptFusion are employed to ensure reproducibility and consistency. For **LERF**, probing is performed using a custom-constructed dataset specifically prepared for this study. For **ConceptFusion**, the evaluation is conducted on the official dataset provided by the ConceptFusion authors.

Model Probing Mechanism. Both LERF and ConceptFusion provide mechanisms for direct querying of their internal 3D language representations. For **LERF**, the optimized network's language field is queried by sampling a dense grid of 3D points, obtaining their CLIP embeddings, and computing the cosine similarity with the query text's embedding. For **ConceptFusion**, the relevancy score for each point in the final 3D point cloud is calculated as the cosine similarity between its stored feature vector and the query text's embedding. This consistent procedure ensures that both models produce comparable 3D relevancy maps for each query.

Query Design and Structure. A carefully designed set of textual queries is used to evaluate the models' ability to understand and localize linguistic concepts within 3D scenes. The queries are structured into four categories of increasing linguistic and visual complexity. The distribution of the number of queries across categories is adjusted according to the nature of the analysis.

- **Category 1: Simple Object Presence (15 queries)**
  These queries test the fundamental ability to identify and localize common, well-defined objects.
  Examples: "a chair," "the television," "a pillow," "the keyboard."

- **Category 2: Fine-grained and Attribute-based (6–7 queries)**
  These queries assess specificity and the models' capacity to ground visual attributes such as color, material, and texture.
  Examples: "a wooden chair," "the blue pillow," "a metal lamp," "the texture on the brick wall."

- **Category 3: Spatial Relationships (6–7 queries)**
  These queries probe compositional understanding and the ability to reason about the relative spatial arrangement of objects.
  Examples: "the trash can under the desk," "a book to the left of the monitor," "the plant on the windowsill," "the object between the two speakers."

- **Category 4: Complex and Abstract Concepts (10 queries)**
  This category explores higher-level reasoning and abstract understanding, divided into two subcategories:
  - **(a) Functional / Affordance-based Queries (5 queries)** — These test the models' understanding of object functionality and affordances.
    Examples: "something you can sit on," "a source of light."
  - **(b) Abstract / Conceptual Queries (5 queries)** — These probe abstract, relational, or scene-level concepts beyond simple object recognition.
    Examples: "the most cluttered area in the room," "a place to put my keys."

This structured probing framework provides a balanced and nuanced evaluation of the linguistic grounding performance of **LERF** and **ConceptFusion** across progressively challenging semantic categories, allowing for a meaningful comparative analysis between the two models.

5 Evaluation Plan

5.1 Quantitative Metrics

3D Intersection over Union (3D IoU): For queries in Categories 1 and 2 that refer to specific, localizable objects, 3D IoU is the gold-standard metric for localization accuracy. The process is as follows:

1. The continuous 3D relevancy map is thresholded at an appropriate level (e.g., 0.5) to produce a binary segmentation of the 3D space into "relevant" and "not relevant" regions.

2. The 3D IoU is calculated as the volume of the intersection between this predicted relevant region and the ground-truth 3D bounding box of the target object, divided by the volume of their union:

$$\text{IoU} = \frac{\text{Volume}(\text{Predicted} \cap \text{GroundTruth})}{\text{Volume}(\text{Predicted} \cup \text{GroundTruth})} \quad (1)$$

Grounding Accuracy (Acc@IoU): Following standard practice in the visual grounding literature, a grounding attempt is considered a "success" if the 3D IoU between the predicted region and the ground-truth object exceeds a certain threshold. We report accuracy at two common thresholds: Acc@0.25 IoU and Acc@0.5 IoU.

Precision and Recall for Abstract Concepts: For queries in Categories 3 and 4, where ground-truth bounding boxes are often ill-defined or non-existent, a different quantitative approach is needed. For a subset of these queries, a sparse set of 3D points will be manually annotated as the "ground-truth relevant set." The evaluation can then be framed as an information retrieval task:

- All points in the scene's point cloud are ranked by their relevancy score from the model.

- Precision is computed as the fraction of retrieved points that are in the ground-truth set.

- Recall is computed as the fraction of ground-truth points that are retrieved at different retrieval depths (k).

This provides a numerical measure of performance even for abstract queries.

5.2 Qualitative Assessment

Quantitative metrics can sometimes be misleading and do not capture the full behavior of a model. Therefore, a thorough qualitative analysis will be conducted.

Visual Inspection of Heatmaps: For a representative subset of queries from all four categories, the generated 3D relevancy heatmaps from LFM, LERF, and ConceptFusion will be rendered side-by-side. These visualizations will be critically examined to assess:

- **Specificity:** How tightly does the relevancy map conform to the target object or region? Are there significant "bleeds" into neighboring objects or the background?
- **Completeness:** Is the entire extent of the relevant object or region highlighted, or does the model only focus on a small, discriminative part?
- **Spurious Activations:** Does the model produce high-relevancy scores in completely irrelevant parts of the scene (false positives)?

Failure Mode Analysis: A systematic analysis of the cases where each model fails will be performed. Failures will be categorized to identify systemic weaknesses. For example:

- Does a model consistently fail on small objects?
- Is it sensitive to occlusion?
- Does it confuse objects with similar colors or shapes?
- Does it fail to parse complex spatial prepositions correctly?

This analysis is crucial for understanding the underlying limitations and behavior of each model.

6 Progress to Mid submission

6.1 Enhancing LERF for General Queries

LERF effectively generates relevancy maps for objects but cannot handle general queries as it only detects object-level relevance. To address this limitation, we developed a two-step pipeline that converts general queries into object descriptions compatible with LERF.

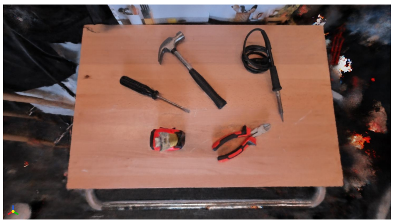
Step 1: Rendering Scenes from NeRF Point Clouds: We render RGB images from NeRF-reconstructed point clouds of the scene using predefined camera poses looking at the scene. This produces images that capture the scene from a controlled viewpoint for subsequent processing.

Figure 3: RGB image rendered from a point cloud

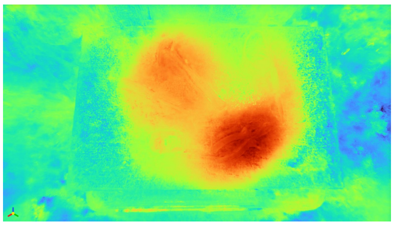
Step 2: Object Description via LLaVA: The rendered images are passed to LLaVA along with a general query. LLaVA outputs concise 2–3 word object descriptions, which LERF can interpret. Example mappings include:

- *The object you would pick to hang a picture, not the one that loosens screws* $\rightarrow$ Hammer
- *Used to bend a wire or remove a small nail* $\rightarrow$ Pliers
- *Which tool can I use to rotate screws and fix them in wall* $\rightarrow$ Screwdriver

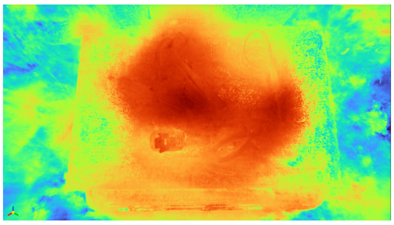
Experimental Example To illustrate this pipeline, we consider a scene containing a hammer, screwdriver, soldering iron, measuring tape, and pliers. All images are rendered for direct comparison.



(a) RGB point cloud image of the scene.



(b) LERF relevancy map for object query "Pliers".



(c) LERF relevancy map for task-based query "Used to bend a wire or remove a small nail".

Figure 4: Comparison of the RGB point cloud and LERF relevancy maps from the same viewpoint. Task-based queries are translated to object queries via LLaVA.

As seen in Figure 4, LERF produces accurate relevancy maps for simple object queries (Fig. 4b). For task-based queries that LERF cannot handle directly (Fig. 4c), LLaVA correctly identifies the target object ("Pliers") from the instruction. LERF then generates a relevancy map over the pliers, successfully grounding complex instructions into localizable objects.

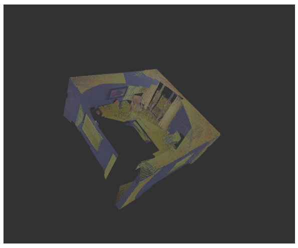
This pipeline extends LERF's capabilities to handle general text instructions and enables evaluation of object localization in task-oriented scenarios.

7 Results/Analysis

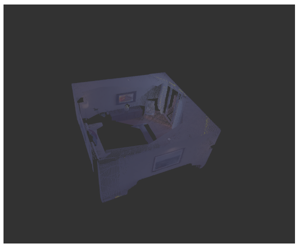
Code The code for this project is available at `https://github.com/codepk37/anlp_group`.

7.1 Probing ConceptFusion for 3D Scene Understanding

To analyze the model's grounding capabilities, we conduct a series of query-based probes on the ICL-NUIM `livingroom1` dataset. Our analysis is structured by query complexity, from simple object presence to abstract reasoning.



(a) Query: "Pot"



(b) Query: "Flower Pot"

Figure 5: Demonstration of semantic ambiguity. The generic query "pot" highlights most of the scene, while the specific query "flower pot" (for the same object) fails entirely.

7.1.1 Simple Object & Attribute Probing

We first test the model's ability to localize simple objects and their attributes. We observe significant ambiguity and a failure to bind attributes correctly.

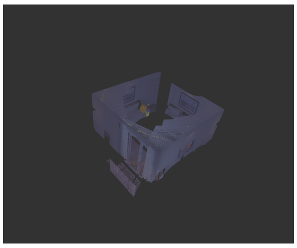
For example, as shown in Figure 5, the query "pot" incorrectly highlights large, unrelated areas of the scene, while the semantically similar query "flower pot" fails to highlight anything. This suggests a strong sensitivity to query phrasing and a lack of robust semantic equivalence.

Furthermore, Figure 6 demonstrates a clear attribute binding failure. While the query "green pillow" (Figure 6a) is correctly localized, the query "blue pillow" (Figure 6b) incorrectly bleeds onto the adjacent pillow. This indicates the model identifies "blue" and "pillow" as co-occurring features in a region but cannot bind them to a specific object instance.

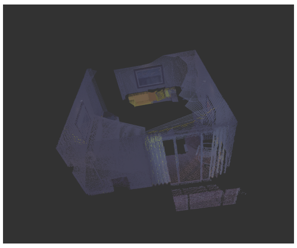
7.1.2 Spatial and Abstract Reasoning

We find that the model struggles significantly as query complexity moves from concrete objects to spatial or abstract relationships.

Spatial Relationships Queries involving spatial relationships (e.g., "beside," "on") consistently failed. As noted in our observations, a query like "chair beside lamp" resulted in highlighting the general surroundings of the *lamp*, not the *chair* qualified by the lamp. This demonstrates a failure to parse the query's relational structure, instead

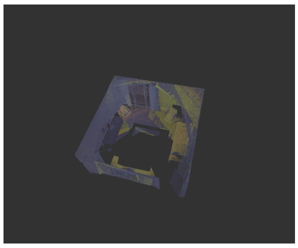


(a) Query: "Green pillow" (Successful)

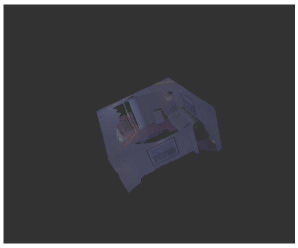


(b) Query: "Blue pillow" (Binding Failure)

Figure 6: Attribute binding failure. The model successfully isolates the "green pillow" but fails to distinguish the "blue pillow" from an adjacent pillow, highlighting both.



(a) Query: "Place to sit" (Success)



(b) Query: "Things symbolizing modern living" (Failure)

Figure 7: Contrasting functional affordance with abstract reasoning. The model can ground "place to sit" in corresponding objects but fails to ground the abstract concept "modern living" to any object.

(a) RGB View 1

(b) RGB View 2

Figure 8: RGB views of the 3D living room scene used for LERF probing.

treating it as a "bag-of-words" and highlighting the region of interest for all mentioned objects.

Abstract vs. Functional Concepts The model's most distinct failure mode is in distinguishing functional affordances from true abstract concepts.

- **Functional Affordance (Success):** A query for "place to sit" (Figure 7a) successfully highlights both the sofas and chairs. This concept is strongly tied to the visual features of objects in the model's pre-training.
- **Abstract Concept (Failure):** In contrast, a query for "things symbolizing modern living" highlights nothing. Even though this query *should* apply to objects like the television (which is found by "Electronic device"), the model is incapable of this symbolic, multi-step reasoning. This result suggests the model's "conceptual" understanding is limited to direct visual-linguistic correlations (affordances, simple categories) and lacks the deeper abstract reasoning required for true scene understanding.

7.2 Analysis of LERF by probing

This section presents a detailed analysis of the LERF model's performance when probed with various query types on a 3D living room scene. The analysis is based on the observed outputs for four distinct categories of queries.

7.2.1 Simple Object Presence Queries

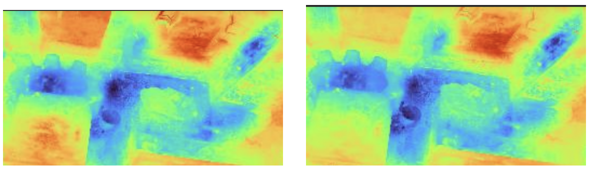
This category tested the model's fundamental ability to identify and locate common objects within the scene.

Analysis: The model demonstrates **mixed and inconsistent performance** in basic object identification.

- **Strengths:** It can successfully and accurately locate common, well-defined objects like `'Sofa'` and `'Chair'`.
- **Precision Sensitivity:** The model shows a clear sensitivity to query precision. For example, the query `'TV'` yielded a more accurate result (highlighting only the television) than the query `'Television'`, which incorrectly included surrounding wall sections. This

suggests the model may be over-indexed on specific keywords. (See Figure 9).

- **Semantic Confusion:** The model struggles with closely related or hierarchical concepts. This is evident in the confusion between `'Flower pot'`, `'Pot'`, and `'Plant'`:
  * `'Flower pot'` highlighted all three pots in the scene, not just the one containing flowers.
  * `'Pot'` highlighted the two pots containing plants but *excluded* the flower pot.
  * `'Plant'` highlighted both plants but also included the flower pot, showing an inability to disentangle distinct, though related, objects.
- **Complete Failure:** The model fails entirely on broad or ambiguous categories. The query `'Table'` resulted in "everything" being highlighted, indicating a total lack of comprehension for this query. Similarly, `'Console'` highlighted the correct object but also "a lot of other things," rendering the result useless.



(a) Query: `'Television'` (Includes walls)

(b) Query: `'TV'` (More precise)

Figure 9: Comparison of precision sensitivity for the queries `'Television'` vs. `'TV'`.

Conclusion: While capable of identifying some simple objects, the model's reliability is compromised by its sensitivity to specific phrasings and its poor handling of semantic ambiguity and broad categories.

7.2.2 Fine-grained and Attribute-based Queries

This category assessed the model's ability to combine object identity with specific attributes, such as color or sub-type.

Analysis: The model's capacity to handle attribute-based queries is **highly unreliable and generally poor.**

- **Attribute Filtering Failure:** The model frequently ignores the attribute entirely. For the query `'Brown chair'` (Figure 10), the model highlighted *all* chairs, completely disregarding the 'Brown' attribute. Likewise, `'Plant in a white pot'` simply highlighted all plants and pots, failing to use 'in a white pot' as a filter.
- **Successful Sub-category Identification:** A notable exception was the query `'Dining chair'`, which correctly isolated the dining chairs from the other accent chair. This suggests the model may have some understanding of *functional sub-types* (e.g., 'dining') but not visual attributes (e.g., 'brown').
- **Object Boundary and Association Errors:** The model struggles to delineate complex objects. `'Dining table'` incorrectly included the dining chairs in its selection, while `'Coffee table'` incorrectly included a separate side table and the flower pot on it. This shows a failure to separate distinct, though nearby, items.

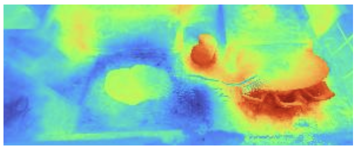


Figure 10: Query: `'Brown chair'`. The model highlighted all chairs, ignoring the 'Brown' attribute.

Conclusion: The model largely fails to use attributes as filters, showing a significant weakness in fine-grained understanding. Its one success in identifying a functional subtype (`'Dining chair'`) stands in sharp contrast to its failure to process simple visual attributes like color.

7.2.3 Spatial Relationships Queries

This category tested the model's understanding of the spatial arrangement of objects relative to one another.

Analysis: The model demonstrates a **very limited and unreliable understanding** of spatial relationships.

- **Ignoring the Relationship:** In most cases, the model appears to default to identifying the *objects* named in the query while ignoring the *relationship* between them.

  * `'Chair beside sofa'` failed, simply highlighting the sofa most and all chairs slightly.
  * `'Plant on the coffee table'` also failed, highlighting both plants in the scene and the flower pot, not just the specific plant on the table.

- **Subject-Object Confusion:** In a partial success, `'Pillow nearest to flower pot'` correctly highlighted the target pillow but also highlighted the flower pot with the same intensity. This shows an inability to distinguish the subject of the query (the pillow) from the relational object (the pot).
- **Ambiguous Success:** The query `'Chair beside dining table'` was successful. However, this query is functionally identical to the `'Dining chair'` query, which also worked. It is highly probable that the model succeeded by identifying the 'Dining chair' sub-type, not by correctly processing the 'beside dining table' spatial command.

Conclusion: The model does not demonstrate a robust ability to parse spatial relationships. It tends to ignore the relational terms and instead identifies the objects mentioned, or it confuses the subject and object of the query.

7.2.4 Complex and Abstract Concepts Queries

This category probed the model's highest-level reasoning, including its understanding of object function (affordance) and abstract ideas.

Functional / Affordance-based Queries Analysis: The model's grasp of object function is **erratic and mostly incorrect.**

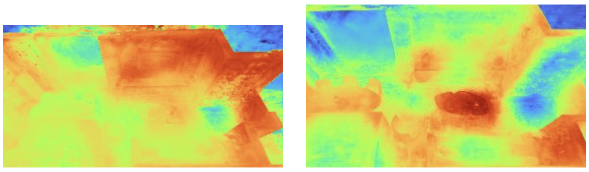
- **Single Success:** The model's only clear success in this entire complex category was `'Object used for dining'`, where it correctly identified the dining table and chairs as a functional set.
- **Total Failure on Broad Affordance:** The query `'Place to sit'` was a complete failure, highlighting "everything except wall" and showing no specific understanding.
- **Misattribution of Function:** The model demonstrated a critical misunderstanding of function. `'Object used for entertainment'` failed to identify the TV and instead highlighted the coffee table, walls, and dining area. `'Objects used for decoration'` was incomplete, identifying the flower pot but missing the obvious sculpture.

Conclusion: The model almost completely fails to connect objects to their functions, with the sole exception of the dining set.

Abstract / Conceptual Queries Analysis: This category revealed the model's **most significant weaknesses and its fragility** to minor query changes.

- **Failure to Map Concepts:** The model consistently fails to map abstract concepts to relevant objects.
  * `'Electronic device for entertainment'` and `'Things symbolizing modern living'` both failed to identify the TV, instead highlighting walls, pillows, or flower pots.
  * `'Objects associated with coziness'` correctly identified the 'sofa' but incorrectly included the 'flower pot', demonstrating flawed association.
- **Extreme Fragility to Phrasing:** The most critical finding is the model's extreme sensitivity to semantically insignificant query changes, such as pluralization. (See Figure 11).
  * `'Object showcasing artwork'` (singular) prompted the model to highlight *walls*.
  * `'Objects showcasing artwork'` (plural) produced a completely different, and still incorrect, result, highlighting the *coffee table* most, along with the flower pot and sculpture.

This inconsistency was also seen with `'Object adding beauty'` (highlighted chairs, pot, sculpture) versus `'Objects adding beauty'` (highlighted pot most, chair, but *missed* the sculpture).



(a) Query: `'Object...'` (Singular)

(b) Query: `'Objects...'` (Plural)

Figure 11: Example of extreme fragility. The queries `'Object showcasing artwork'` (left) and `'Objects showcasing artwork'` (right) produce completely different and incorrect results, highlighting walls and a coffee table, respectively.

Conclusion: The model has no discernible grasp of abstract concepts. Furthermore, the wild variations in output between singular and plural versions of the same query (e.g., `'Object'` vs. `'Objects'`) demonstrate that the model is not generalizing based on semantic *intent*. Instead, it appears to be highly brittle and over-fitted to specific, arbitrary query phrasings, making it unsuitable for reliable abstract probing.

7.3 Comparative Analysis of ConceptFusion and LERF

A direct comparison of the probing results for ConceptFusion (Section 8.1) and LERF (Section 8.2) reveals distinct and informative failure modes for each architecture, even when they fail on similar query categories.

Simple Object Ambiguity and Precision Both models demonstrate a significant "brittleness" to query phrasing and semantic ambiguity. This is best illustrated by the 'Pot' vs. 'Flower Pot' queries. For ConceptFusion, 'Pot' highlighted "everything" while 'Flower Pot' highlighted "nothing". LERF showed similar, though distinct, confusion, highlighting all pots for 'Flower pot' and only some for 'Pot'. Both models also showed precision sensitivity: LERF performed better with 'TV' than 'Television' , while ConceptFusion produced different maps for 'Curtain' (with noise) and 'Curtains' (cleanly). This suggests neither model possesses robust semantic equivalence for simple objects.

Attribute Binding The models show a key architectural difference in attribute binding. LERF's primary failure is **ignoring the attribute**; for 'Brown chair', it highlighted all chairs, completely disregarding the color 'Brown'. In contrast, ConceptFusion's failure is one of **imprecise binding**, or "bleeding"; for 'Blue pillow', it correctly identified the target but also highlighted the adjacent pillow. Notably, LERF succeeded at a *functional* subtype ('Dining chair'), suggesting its understanding is stronger for functional categories than for simple visual attributes, where ConceptFusion ('Green pillow']) showed some success.

Spatial Relationships This category represents a universal failure for both models. They both fundamentally fail to parse compositional spatial language, degrading into a "bag-of-words" model that simply highlights all mentioned objects. For LERF, 'Chair beside sofa' highlighted the sofa most and all chairs slightly , and 'Duck above Rubik's cube' highlighted both objects equally, ignoring "above" . ConceptFusion performed identically; 'Sofa beside chair' highlighted "both sofas and chairs". This demonstrates that true spatial reasoning is absent in both.

Abstract and Functional Concepts The most significant divergence between the models is in abstract reasoning.

- **Functional Affordance:** ConceptFusion demonstrated a clear, successful understanding of simple affordances. The query 'Place to sit' correctly highlighted all sofas and chairs. LERF failed this exact same query catastrophically, highlighting "everything except wall".
- **Abstract Concepts:** Both models failed to ground true abstract concepts, such as 'Things symbolizing modern living'.
- **Query Fragility:** A critical finding unique to LERF was its *extreme fragility* to minor phrasing, such as singular vs. plural. The query 'Object showcasing

artwork' (highlighting walls) and 'Objects showcasing artwork' (highlighting the coffee table) produced "completely different and incorrect results". This suggests LERF's abstract reasoning is not just weak but highly brittle and "overfitted to specific, arbitrary query phrasings", a flaw not observed to the same degree in the ConceptFusion probes.

In summary, while both models fail at complex spatial and abstract reasoning, ConceptFusion shows a superior ability to ground functional affordances. Conversely, LERF's failures are marked by a complete inability to filter by visual attributes and an extreme, disqualifying sensitivity to semantically minor changes in query phrasing.

References

Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. 2021. Emerging properties in self-supervised vision transformers. *Preprint*, arXiv:2104.14294.

Shivam Chandhok. 2024. Scenegpt: A language model for 3d scene understanding. *Preprint*, arXiv:2408.06926.

Maxim Eraafat. 2023. Blendernerf: Easy nerf synthetic dataset creation within blender. https://github.com/maximeraafat/BlenderNeRF. Accessed: 2025-10-02.

Krishna Murthy Jatavallabhula, Alihusein Kuwajerwala, Qiao Gu, Mohd Omama, Tao Chen, Alaa Maalouf, Shuang Li, Ganesh Iyer, Soroush Saryazdi, Nikhil Keetha, Ayush Tewari, Joshua B. Tenenbaum, Celso Miguel de Melo, Madhava Krishna, Liam Paull, Florian Shkurti, and Antonio Torralba. 2023. Conceptfusion: Open-set multimodal 3d mapping. *Preprint*, arXiv:2302.07241.

Justin Kerr, Chung Min Kim, Ken Goldberg, Angjoo Kanazawa, and Matthew Tancik. 2023. Lerf: Language embedded radiance fields. *Preprint*, arXiv:2303.09553.

Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. 2023. Visual instruction tuning. *Preprint*, arXiv:2304.08485.

Ben Mildenhall, Pratul P. Srinivasan, Matthew Tancik, Jonathan T. Barron, Ravi Ramamoorthi, and Ren Ng. 2020. Nerf: Representing scenes as neural radiance fields for view synthesis. *Preprint*, arXiv:2003.08934.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. 2021. Learning transferable visual models from natural language supervision. *Preprint*, arXiv:2103.00020.

A Additional Experiments: LERF Limitations

We conducted additional qualitative experiments to probe the reasoning capabilities of LERF. The examples below highlight cases where the model struggles with queries involving spatial or comparative reasoning.

Figure 12: Scene used for LERF text-query experiments.

Scene Setup: The reference scene used for all text–query experiments.

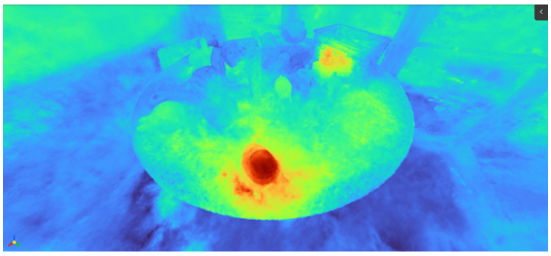


Figure 13: LERF correctly detects both duck instances.

Query: Duck
Observation: LERF correctly highlights both duck instances (in red), demonstrating instance-level object recognition.

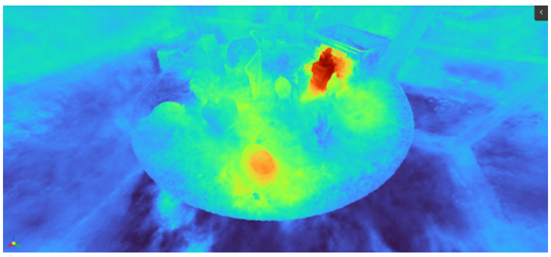


Figure 14: Failure case: LERF cannot reason about spatial relationships.

Query: Duck above Rubik's cube
Observation: Both ducks and the Rubik's cube are highlighted. LERF fails to capture the spatial relationship above; it detects object presence but not relative arrangement.

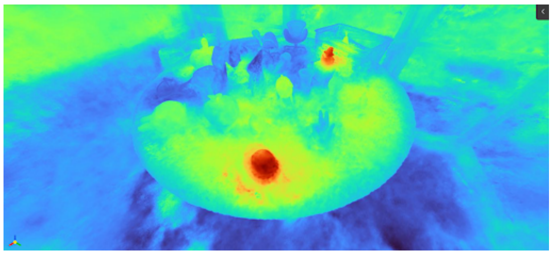


Figure 15: Failure case: LERF cannot reason over comparative size attributes.

Query: Bigger duck
Observation: Both ducks are highlighted equally without distinction. LERF does not reason about relative size attributes between objects.

B Additional analysis by probing of ConceptFusion and LERF

All types of queries are analyzed in detailed in this ablation. The full ablation study can be found here: Ablation Study Link

Future Work: We will also try to address these limitations and improve the model's handling of spatial relationships and comparative attributes.